

Agreement Through Calibrated Conversation

When Human and AI Predictions Can Correct Each Other

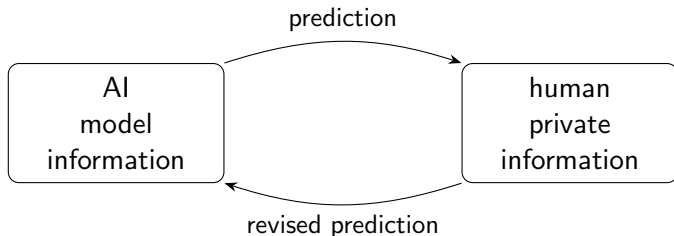
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Fall 2026

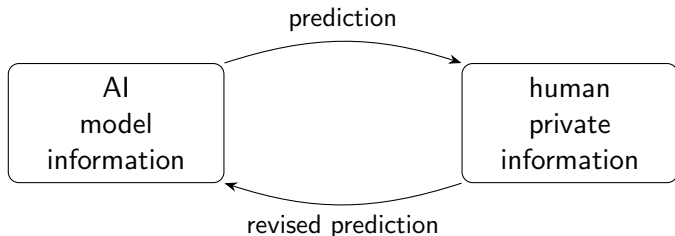
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Both want an accurate probability for the same event, such as a safety failure. Neither party's initial estimate should automatically settle the question.

The alignment question

Lecture 7 made a calibrated forecast a one-way decision interface. When the interface runs in both directions, can exchanging predictions resolve disagreement and improve accuracy?

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- ▶ We seek fast approximate agreement. Conversation calibration retains a learnable consequence of Bayesian rationality without requiring exact Bayesian posteriors.
- ▶ Calibrated replies preserve the openings' aggregate accuracy. Continued disagreement yields further improvement, up to calibration error—so sustained disagreements are rare.
- ▶ Approachability enforces approximate conversation calibration from repeated outcome feedback.

Private Information and Bayesian Probabilities

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Fix an event $E \subseteq \Omega$. Its *posterior probability* is

$$p_i(\omega) := \mu(E \mid \mathcal{I}_i(\omega)) = \frac{\mu(E \cap \mathcal{I}_i(\omega))}{\mu(\mathcal{I}_i(\omega))}.$$

Here E stays fixed: changing ω changes the information available to the agent, not the event whose probability we ask for.

Different Evidence Can Justify Different Probabilities

Two bits X_A, X_H are independent and uniform. The AI sees X_A , the human sees X_H , and the event is $E = \{X_A = X_H = 1\}$.

state (X_A, X_H)	prior	$\mathbf{1}_E$	AI posterior	human posterior
(1, 1)	1/4	1	1/2	1/2
(1, 0)	1/4	0	1/2	0
(0, 1)	1/4	0	0	1/2
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At state (1, 0), both agents reason correctly but initially disagree. When the human reports probability zero, the AI learns that $X_H = 0$ and updates its probability to zero.

Aumann's theorem concerns probabilities that are *common knowledge*, not arbitrary probabilities computed from separate evidence.

Knowledge and Common Knowledge

Agent i *knows* an event F at state ω if every state it considers possible belongs to F :

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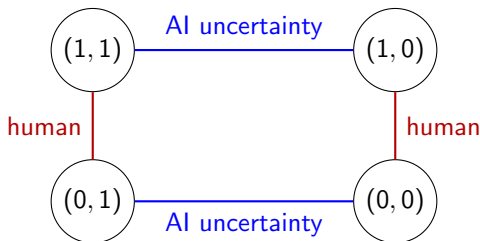
Common knowledge requires more than both agents knowing F : each knows it, each knows that the other knows it, and so on. There is a finite way to express all these requirements. Connect two states whenever either agent cannot distinguish them. Let $D(\omega)$ be the connected component containing ω . Then F is common knowledge at ω precisely when

$$\boxed{D(\omega) \subseteq F.}$$

Why: “H knows A knows F ” says every $\mathcal{I}_A$ -cell touching $\mathcal{I}_H(\omega)$ lies in F . Each further level alternates partitions once more, and the levels together sweep out exactly the connected component.

Common Knowledge Joins Whole Information Cells

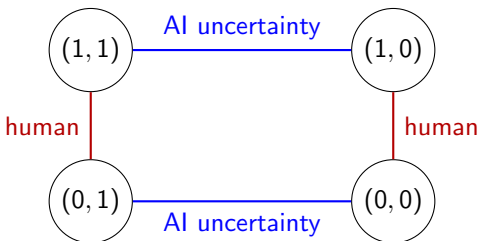
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In general, if one state of an information cell belongs to $D(\omega)$, every state of that cell does. Thus $D(\omega)$ is a union of whole cells of *either* agent's partition.

Aumann's Agreement Theorem

Theorem (Aumann, 1976)

*Two Bayesian agents share a finite, full-support common prior μ .
If at state ω it is common knowledge that their posterior
probabilities of the same event E are*

$$p_A = \alpha \quad \text{and} \quad p_H = \beta,$$

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Recall what the state argument means:

$$p_i(\omega') = \mu(E \mid \mathcal{I}_i(\omega')).$$

This is agent i 's probability of the *fixed event* E when the state is ω' , not its probability of the state ω' itself.

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The common-knowledge claim concerns the agents' probabilities of E , not whether E occurred. The cell D may contain states both inside and outside E .

Both Probabilities Average to the Same Number

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$$\begin{aligned}\mu(E \mid D) &= \sum_{I \in \mathcal{I}_A: I \subseteq D} \mu(I \mid D) \mu(E \mid I) \\ &= \sum_{I \in \mathcal{I}_A: I \subseteq D} \mu(I \mid D) \alpha \\ &= \alpha.\end{aligned}$$

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The same calculation using the human's cells gives $\mu(E \mid D) = \beta$.

Hence $\alpha = \beta$. □

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Hearing a report can change your posterior, so announcing beliefs once need not settle the matter. Can continued exchange make the *current* posterior values common knowledge?

The Transcript Determines the Remaining States

Let $\Omega_j \subseteq \Omega$ be the states consistent with the first j public announcements, starting from $\Omega_0 = \Omega$. The current posterior of agent i at $\omega \in \Omega_j$ is

$$p_i^{(j)}(\omega) := \mu(E \mid \mathcal{I}_i(\omega) \cap \Omega_j).$$

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If the agent announces v , this set is refined:

$$\Omega_{j+1} = \{\omega' \in \Omega_j : p_i^{(j)}(\omega') = v\} \subseteq \Omega_j.$$

The same transcript narrows the possibilities for both agents, while each still has its own private information cell.

At Stabilization, Aumann Forces Agreement

The public sets form a decreasing sequence

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that always contains the actual state. Since Ω is finite and each nontrivial update removes at least one state, the sequence eventually terminates at a nonempty set Ω_* .

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From then on, each agent's announcement removes no state. So for some public values α, β ,

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After the public announcements, the state space is Ω_* , the common prior is $\mu(\cdot \mid \Omega_*)$, and the information cells are $\mathcal{I}_i(\omega) \cap \Omega_*$. Both current posterior values hold at every state reachable through either agent's uncertainty. They are therefore *common knowledge* in the updated model, and Aumann's theorem gives

$$\boxed{\alpha = \beta.}$$

We Want Fast Approximate Agreement

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The quantitative goal

For a tolerance $\varepsilon > 0$, can consecutive reports become ε -close after few messages on most conversations, while preserving accuracy—with a round bound that does not depend on $|\Omega|$?

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We will keep the alternating communication pattern and replace exact Bayesian updating with a calibration condition learned from outcome feedback: a *tractable relaxation of Bayesian rationality* that does not require a common prior.

Repeated Conversations with Outcome Feedback

We now repeat the interaction on days $t = 1, \dots, T$. An arbitrary binary outcome $y_t \in \{0, 1\}$ is fixed before each conversation; each party has private information about it. Reports come from a finite grid $\mathcal{P} \subset [0, 1]$ of size $M = |\mathcal{P}|$.

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Openings. In the learned protocol, allow two unrestricted opening reports: the AI states p_t^1 and the human states p_t^2 , each from its own information.

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Fix a tolerance $\varepsilon \in (0, 1)$ and a cap of $R \geq 3$ messages. Stop at the first reply round $k \geq 3$ with $|p_t^k - p_t^{k-1}| \leq \varepsilon$; if agreement has not occurred, stop at the cap. Then reveal y_t , which can inform future days.

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We no longer assume a common prior or exact Bayesian updates. We will instead impose calibration conditions on the *replies* of both speakers.

Calibration is not Enough

Suppose half the days have each outcome. The human sees y_t exactly and reports it. The AI always reports $1/2$, ignoring every human message.

fraction of days	outcome	AI reports	human reports
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Every consecutive pair of reports differs by $1/2$. For $\varepsilon < 1/2$, no conversation reaches agreement, however large the cap.

Calibration alone has not required the AI to use the message it just received.

The Previous Message Exposes the Ignored Information

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This is the context-dependent calibration, as in Lecture 9.

Which Conversations Reach k Rounds?

Let S_k be the set of days in which conversation reaches message k . Every conversation involves both openings and the first reply, so $S_1 = S_2 = S_3 = \{1, \dots, T\}$. For $k \geq 3$,

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Recall that we allow at most R messages per conversation. The fraction still that have not reached agreement at this cap is

$$\frac{|S_{R+1}|}{T}.$$

Calibration Within Each Pair of Reports

At reply round $k \geq 3$, for previous report q and reply p , define

$$C_{k,q,p} := \{t \in S_k : p_t^{k-1} = q, p_t^k = p\}.$$

Its accumulated prediction error is

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Exact calibration requires $r_{k,q,p} = 0$ for every pair. On each nonempty cell, this says that the outcomes average to the reply p . At the AI's first reply, impose the same check against the AI's opening:

$$s_{q,p} := \sum_{t: p_t^1=q, p_t^3=p} (y_t - p) = 0 \quad \text{for every } q, p.$$

Thus $r_{3,q,p}$ conditions on the human's opening; $s_{q,p}$ conditions on the AI's.

Why Check the AI's First Reply Against Both Openings?

The extra check $s_{q,p} = 0$ compares the AI's first reply with the AI's own opening. What would go wrong without it?

Suppose half the days have each outcome. The AI's opening is perfect, $p_t^1 = y_t$, while the human's opening is $p_t^2 = 1/2$.

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If the AI replies $p_t^3 = 1/2$ every day, it passes the previous-message check: the single $(p^2, p^3) = (1/2, 1/2)$ cell has outcome mean $1/2$. The conversation immediately ends with report $1/2$, discarding the AI's perfect opening.

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The extra check rejects this reply. Within a (p^1, p^3) cell, the outcome mean is p^1 —either 0 or 1, not the reply $1/2$.

This additional check lets us prove that the conversation is only accuracy improving.

Approximate Calibration Allows Some Residual Error

At reply round k , measure the total error across cells by

$$\Lambda_k := \sum_{q,p} |r_{k,q,p}| + \mathbf{1}\{k = 3\} \sum_{q,p} |s_{q,p}|.$$

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We will make these finitely many residuals small using outcome feedback, without computing posteriors over Ω or conditioning on the responder's full private information.

Conversation Calibration Forces Fast Agreement

Theorem

Suppose $\Lambda_k \leq \gamma T$ for each reply round $k = 3, \dots, R$. Then the fraction of conversations that fail to reach ϵ -agreement satisfies

$$\frac{|S_{R+1}|}{T} \leq \frac{1}{(R-2)\epsilon^2} + \frac{2\gamma}{\epsilon^2}.$$

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For $\delta \in (0, 1)$, exact calibration ensures that at least a $1 - \delta$ fraction of conversations reach ϵ -agreement within

$$R = 2 + \left\lceil \frac{1}{\delta \epsilon^2} \right\rceil$$

messages: two openings and $\lceil 1/(\delta \epsilon^2) \rceil$ replies.

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The proof's idea: substantial disagreement forces an improvement in total squared prediction error across days. There is only a bounded amount of error to remove.

One Bounded Error Measure Tracks All the Conversations

For analysis, $\bar{p}_t^k = p_t^k$ if day t reaches round k . If its conversation ends at round h , keep using that final report:

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No further messages are sent; this only extends the notation beyond stopping.

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No further messages are sent; this only extends the notation beyond stopping.

Our measure of prediction error is

$$\mathcal{E}_k := \sum_{t=1}^T (y_t - \bar{p}_t^k)^2, \quad 0 \leq \mathcal{E}_k \leq T.$$

For a binary outcome, $(y - p)^2$ is also called the *Brier loss*. So $\mathcal{E}_1$ and $\mathcal{E}_2$ are the total errors of the two openings.

One Bounded Error Measure Tracks All the Conversations

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A stopped conversation contributes the same squared error at every later round. Thus only days in S_k can change $\mathcal{E}_{k-1}$ to $\mathcal{E}_k$.

A Squared-Error Identity Measures the Gain from a Reply

On one active day, write f for a report we compare against, p for the reply, and y for the outcome. Since $y - f = (y - p) + (p - f)$,

$$\begin{aligned}(y - f)^2 - (y - p)^2 &= ((y - p) + (p - f))^2 - (y - p)^2 \\ &= (p - f)^2 + 2(y - p)(p - f).\end{aligned}$$

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The first term is the squared change from f to the reply. The second can have either sign on an individual day.

To turn a large change into an accuracy improvement, we must control that second term after averaging over days.

Pairwise Calibration Controls the Cross Term

First compare the reply with the previous report: $f_t = p_t^{k-1}$.
Within $C_{k,q,p}$, the two reports are fixed at q and p , so

$$\sum_{t \in C_{k,q,p}} 2(y_t - p)(p - q) = 2(p - q) \underbrace{\sum_{t \in C_{k,q,p}} (y_t - p)}_{r_{k,q,p}}.$$

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Summing over cells and using $|p - q| \leq 1$ gives

$$\sum_{t \in S_k} 2(y_t - p_t^k)(p_t^k - f_t) \geq -2 \sum_{q,p} |r_{k,q,p}| \geq -2\Lambda_k.$$

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At round 3, the same calculation with $f_t = p_t^1$ uses $s_{q,p}$ instead of $r_{3,q,p}$. Its cross term is also at least $-2\Lambda_3$.

Unresolved Conversations Force Accuracy Progress

Only active days change their report. For $k \geq 4$, take $f = p^{k-1}$ and sum the identity over S_k :

$$\mathcal{E}_{k-1} - \mathcal{E}_k \geq \sum_{t \in S_k} (p_t^k - p_t^{k-1})^2 - 2\Lambda_k.$$

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Every day in S_{k+1} has $|p_t^k - p_t^{k-1}| > \varepsilon$. The other squared changes are nonnegative, so

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At the first reply:

$$\mathcal{E}_2 - \mathcal{E}_3 \geq \varepsilon^2 |S_4| - 2\Lambda_3, \quad \mathcal{E}_1 - \mathcal{E}_3 \geq -2\Lambda_3.$$

With conversation calibration, a large change cannot merely move error around: it reduces the total squared error.

The Bounded Error Limits the Number of Messages

Sum the progress bounds from $k = 3$ through R ; the error terms telescope:

$$\begin{aligned}\varepsilon^2 \sum_{k=3}^R |S_{k+1}| &\leq \sum_{k=3}^R (\mathcal{E}_{k-1} - \mathcal{E}_k) + 2 \sum_{k=3}^R \Lambda_k \\ &= \mathcal{E}_2 - \mathcal{E}_R + 2 \sum_{k=3}^R \Lambda_k \leq T + 2 \sum_{k=3}^R \Lambda_k.\end{aligned}$$

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The active sets are nested, so each of the $R - 2$ terms $|S_{k+1}|$ is at least $|S_{R+1}|$. Hence, dividing by $\varepsilon^2(R - 2)T$:

$$\boxed{\frac{|S_{R+1}|}{T} \leq \frac{1}{(R - 2)\varepsilon^2} + \frac{2 \sum_{k=3}^R \Lambda_k}{T(R - 2)\varepsilon^2}.$$

If every $\Lambda_k \leq \gamma T$, this is the theorem. □

Accuracy Relative to Both Opening Predictions

On each day, count the reply rounds $k = 4, \dots, R$ that are reached and leave $|p_t^k - p_t^{k-1}| > \varepsilon$. Let $\bar{N}$ be this count averaged over all T days:

$$\bar{N} := \frac{1}{T} \sum_{k=4}^R |S_{k+1}|.$$

Rounds 1, 2, 3 and any reply that achieves agreement are excluded; all other sent replies count, including at the cap.

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With exact calibration, the final reports improve on the better opening's average squared error by at least $\varepsilon^2 \bar{N}$.

Next we construct replies with small calibration residuals.

What We Assume About the Human

The human plays two different roles in what follows.

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The openings remain arbitrary: each party states whatever its own information supports. Calibration constrains only how replies respond.

One Forecaster Learns Each Position in the Conversation

For each reply round $k = 3, \dots, R$, run a separate procedure across days. On a day that reaches round k :

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1. Observe the previous message and, at round 3, also the AI's opening.
2. Choose a distribution over reports, then draw and send p_t^k .
3. After the conversation, observe y_t and update this procedure.

The outcome is fixed before the day's conversation and all its random draws. It cannot be chosen in response to a sampled report.

Private information may help choose among distributions satisfying the forecasting rule below.

Each Pairwise Check Supplies One Block of Coordinates

Fix reply round k . For an earlier report q , candidate reply p , and outcome y , define the M^2 -coordinate vector

$$v(q, p, y)_{q', p'} := \mathbf{1}\{q = q', p = p'\}(y - p),$$

where $q', p' \in \mathcal{P}$. Only one coordinate can be nonzero.

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where $q', p' \in \mathcal{P}$. Only one coordinate can be nonzero.

On active day t , the complete vector is

$$g_t(p, y) := \begin{cases} (v(p_t^2, p, y), v(p_t^1, p, y)), & k = 3, \\ v(p_t^{k-1}, p, y), & k \geq 4. \end{cases}$$

Set $g_t = 0$ on inactive days. There are at most $2M^2$ coordinates, with at most two nonzero, so $\|g_t(p, y)\|_2 \leq \sqrt{2}$.

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The average $\bar{g}_T = T^{-1} \sum_{t \in S_k} g_t(p_t^k, y_t)$ consists of $r_{k,q,p}/T$, with $s_{q,p}/T$ appended at round 3. Thus

$$\|\bar{g}_T\|_1 = \Lambda_k / T.$$

The Calibration Target Is Response Satisfiable

Choose the finite grid so that every $z \in [0, 1]$ is within η of a grid report. Our target is the closed convex set

$$\mathcal{K}_\eta := \{u : \|u\|_1 \leq 2\eta\} \quad \text{in } \mathbb{R}^{2M^2} \text{ at round 3 and } \mathbb{R}^{M^2} \text{ afterwards.}$$

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We still use Lecture 8's *Euclidean distance* to this target.

Fix the earlier report(s) and any outcome distribution with mean z . Choose p with $|p - z| \leq \eta$. In each block, the expected vector has just one possibly nonzero entry:

$$\mathbb{E}[v(q, p, y)]_{q', p'} = \mathbf{1}\{q = q', p = p'\}(z - p).$$

There are at most two blocks. Hence $\|\mathbb{E}[g_t(p, y)]\|_1 \leq 2\eta$, which puts the expected vector in $\mathcal{K}_\eta$. Inactive days give zero.

Blackwell Supplies the Online Forecasting Rule

The response above was allowed to know the outcome distribution. Lecture 8's minimax argument turns those hypothetical responses into one rule that works against either outcome.

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Initialize $\bar{g}_0 = 0$. After observing the earlier report(s), project $\bar{g}_{t-1}$ onto $\mathcal{K}_\eta$, obtaining c . Set $w = \bar{g}_{t-1} - c$. Choose a distribution ν_t over reports such that

$$\mathbb{E}_{p \sim \nu_t}[\langle w, g_t(p, y) - c \rangle] \leq 0 \quad \text{for both } y \in \{0, 1\}.$$

Response satisfiability guarantees that such a distribution exists. Finding it is a linear feasibility problem in M report probabilities, with one constraint for each of the two outcomes.

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Draw and send $p \sim \nu_t$. When y_t is revealed, add the observed vector to the running average. Each round- k procedure maintains its own average.

The Distance Bound Controls Calibration Error

Since $0 \in \mathcal{K}_\eta$ and $\|g_t\|_2 \leq \sqrt{2}$, Lecture 8 gives

$$\mathbb{E}[\text{dist}(\bar{g}_T, \mathcal{K}_\eta)^2] \leq \frac{8}{T}.$$

Recall $\Lambda_k/T = \|\bar{g}_T\|_1$, the sum of absolute residual coordinates.
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$$\begin{aligned} \frac{\Lambda_k}{T} &\leq \|c\|_1 + \|\bar{g}_T - c\|_1 \\ &\leq 2\eta + \sqrt{2}M\|\bar{g}_T - c\|_2. \end{aligned}$$

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Taking expectations and applying Cauchy–Schwarz once more gives

$$\begin{aligned} \mathbb{E}[\Lambda_k/T] &\leq 2\eta + \sqrt{2}M\sqrt{\mathbb{E}[\text{dist}(\bar{g}_T, \mathcal{K}_\eta)^2]} \\ &\leq 2\eta + \frac{4M}{\sqrt{T}}. \end{aligned}$$

We use this one bound for every reply round.

Learned Calibration Gives Short Conversations

Our deterministic proof established

$$\frac{|S_{R+1}|}{T} \leq \frac{1}{(R-2)\varepsilon^2} + \frac{2\sum_{k=3}^R \Lambda_k}{T(R-2)\varepsilon^2}.$$

Take expectations and use the bound on each $\mathbb{E}[\Lambda_k/T]$:

$$\boxed{\mathbb{E}\left[\frac{|S_{R+1}|}{T}\right] \leq \frac{1}{(R-2)\varepsilon^2} + \frac{4\eta}{\varepsilon^2} + \frac{8M}{\varepsilon^2\sqrt{T}}.}$$

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For any desired tolerance and unresolved fraction, choose enough replies, a fine enough grid, and enough days of feedback. A uniform grid has $M = O(1/\eta)$, so all the required resources are polynomial in the inverse tolerances.

The expectation averages the protocol's random draws. The conclusion controls the fraction of these T conversations that remain unresolved, not every day or a later deployment period.

Learning Preserves the Accuracy Gain from Conversation

Recall that $\bar{N}$ averages, across days, the number of reply rounds after round 3 that still leave consecutive reports more than ε apart.

We proved

$$\frac{\min\{\mathcal{E}_1, \mathcal{E}_2\} - \mathcal{E}_R}{T} \geq \varepsilon^2 \bar{N} - \frac{2}{T} \sum_{k=3}^R \Lambda_k.$$

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Using $\mathbb{E}[\Lambda_k/T] \leq 2\eta + 4M/\sqrt{T}$ for each of the $R - 2$ reply rounds gives

$$\mathbb{E}\left[\frac{\min\{\mathcal{E}_1, \mathcal{E}_2\} - \mathcal{E}_R}{T}\right] \geq \varepsilon^2 \mathbb{E}[\bar{N}] - 2(R - 2) \left(2\eta + \frac{4M}{\sqrt{T}}\right).$$

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The final reports match the better opening's aggregate accuracy up to the displayed allowance. They are strictly better when the progress from these reply rounds with continued disagreement exceeds that allowance.

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- ▶ Conversation calibration is a tractable relaxation of Bayesian rationality: replies are reliable in the context of the reports they answer.
- ▶ Conversation protects the better opening's aggregate accuracy, up to calibration error. Further unresolved exchanges yield additional accuracy improvement, limiting how long disagreement can persist.
- ▶ Next: a shared decision may require much less agreement than matching numerical probabilities. Can the parties exchange actions instead?

Thanks!

See you next class!

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